



Object Oriented Programming

Practice Problem

Pointers (30 Questions)

Basic (10 Questions)

1. **Pointer Basics:** Declare an integer pointer, assign a value, print value using both pointer and variable.
2. **Pointer Arithmetic:** Perform ++, --, +, - operations on an integer pointer and observe changes.
3. **Pointer to an Array:** Access an integer array using pointers instead of indexing.
4. **Swapping Two Numbers Using Pointers:** Swap two numbers using pointer references.
5. **Reverse an Array Using Pointers:** Reverse a static integer array using a pointer.
6. **Find Sum of Array Using Pointers:** Compute sum using only pointer arithmetic.
7. **Function Pointers (Addition):** Create a function to add two numbers using function pointers.
8. **Find Maximum Value in an Array Using Pointers:** Use a pointer to find the max value in an array.
9. **Sort an Array Using Pointers:** Implement Bubble Sort using pointer-based swapping.

10. **Reverse a String Using Pointers:** Reverse a string using character pointers.

Medium (10 Questions)

11. **Count Vowels Using Pointers:** Traverse a character array using pointers and count vowels.

12. **Copy String Without Using strcpy():** Copy one string to another using pointers.

13. **Find Frequency of an Element in an Array Using Pointers:** Count occurrences of an integer in an array using pointers.

14. **Sorting Strings Using Pointers:** Sort an array of strings using pointer manipulation.

15. **Passing Arrays to Functions Using Pointers:** Implement function `sum(int *arr, int size)` to find sum of elements.

16. **Pointer to Pointer Usage:** Demonstrate use of double pointers (`**ptr`).

17. **Find Unique Elements in an Array:** Find and print unique numbers in an array using pointers.

18. **Pointer-Based Matrix Addition:** Add two matrices using only pointer dereferencing.

19. **Implement a Simple Calculator Using Function Pointers:** Perform `+`, `-`, `*`, `/` operations using function pointers.

20. **Dynamically Allocate a String Using Pointers:** Create a string dynamically and modify its characters using pointers.

Advanced (10 Questions)

21. **Pointer-Based Prime Number Finder:** Find all prime numbers in a given range using pointers.

22. **Concatenate Two Strings Using Pointers:** Implement `strcat()` functionality with pointers.

23. **Implement strrev() Using Pointers:** Reverse a string without using built-in functions.

24. **Function Returning a Pointer to an Array:** Create a function that returns a pointer to an integer array.

25. **Replace Vowels in a String Using Pointers:** Replace vowels with '*' in a dynamically allocated string.
26. **Merge Two Sorted Arrays Using Pointers:** Merge two sorted arrays using pointers.
27. **Find Palindromes in a String Using Pointers:** Check if a word in a sentence is a palindrome.
28. **Transpose a Matrix Using Pointers:** Find transpose using pointer-based traversal.
29. **Memory Leak Example and Fix:** Demonstrate a memory leak and fix it using proper delete.
30. **Pointer-Based Substring Extraction:** Extract a substring from a larger string using only pointers.